

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	UNM293SmartStudyAI
Project Title	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform creative storytelling and comic generation using Google Gemini multimodal models, boosting creative efficiency, narrative coherence, and visual engagement. It tackles traditional content bottlenecks, promising intuitive prompt-to-comic generation, customizable artistic panels, and richer creator workflows. Key features include Gemini-powered dynamic narrative scripting, dialogue synthesis, character consistency management, and real-time multi-panel comic layout rendering.

Project Overview	
Objective	The primary objective is to revolutionize the visual comic creation process by implementing advanced Google Gemini multimodal generative models, ensuring instant narrative synthesis, visual-script alignment, and rich illustrated storyboards.
Scope	The project comprehensively addresses and enhances the digital comic creation pipeline, incorporating generative AI for scriptwriting, sequential panel description, character design adherence, and automated comic layout assembly.
Problem Statement	
Description	Traditional comic and graphic story creation requires extensive artistic expertise, high manual drafting time, and disjointed creative toolchains, significantly slowing down independent writers, educators, and indie creators from visualizing dynamic storyboards.
Impact	Solving these challenges democratizes graphic novel creation, reduces storyboard turnaround time from days to minutes, and provides an accessible, intuitive platform for writers, educators, and content creators to turn raw ideas into finished comic panels.
Proposed Solution	
Approach	Employing multimodal Gemini models alongside visual generation pipelines to synthesize interactive story beats, dialogue captions, panel composition prompts, and cohesive multi-panel comic strips from user text prompts.
Key Features	- Implementation of Google Gemini multimodal models for contextual story scripting, character dialogue, and sequential scene structuring. - Real-time prompt-driven panel and storyboard layout generation. - Consistent character attribute tracking and customizable art styles across panels. - Exportable high-resolution digital comic book formats (PDF, PNG, Web Comic).

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		

Computing Resources	CPU/GPU specifications, number of cores	T4 / A100 GPU (Cloud GPU Instance)
Memory	RAM specifications	16 GB RAM
Storage	Disk space for data, models, and logs	512 GB SSD / Cloud Storage
Software		
Frameworks	Application & AI frameworks	Flask / FastAPI, Google GenAI SDK
Libraries	Additional libraries	google-generativeai, Pillow (PIL), requests, pydantic, reportlab
Development Environment	IDE / Platform	VS Code, Google AI Studio, Jupyter Notebook
Data / APIs		
Data & Services	APIs, prompt templates, comic schemas	Google Gemini 1.5 Pro / Flash API, Imagen / Visual Diffusion APIs, JSON Comic Schemas